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Bayesian Statistics and Data Analysis

Lecture 4

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Thanks to Aki Vehtari, Aalto University

- Introduction
 - Computational aspects
- Bayesian Computation
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Additional material



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Notation

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importance sampling

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- In this lecture, generic $p(\theta)$ is used instead of $p(\theta|y)$



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- In this lecture, generic $p(\theta)$ is used instead of $p(\theta|y)$
- An **unnormalized density** is denoted by $q(\cdot)$
 - Its normalizing constant must satisfy $Z = \int q(\theta) d\theta$, with $0 < Z < \infty$
 - The normalized density is $p(\theta) = q(\theta)/Z$, equivalently $q(\theta) \propto p(\theta)$



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 - The normalized density is $p(\theta) = q(\theta)/Z$, equivalently $q(\theta) \propto p(\theta)$
- **Proposal** distribution is denoted by $g(\cdot)$



- Very small positive numbers may be rounded to zero in floating-point calculations
 - closest value to zero is $\approx 2.2 \cdot 10^{-308}$
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` $\rightarrow 0$ (underflow)
(See log densities on the next slide)

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- Probabilities extremely close to 1 may be rounded to 1
 - Ratio of girl and boy babies
 - `pbeta(0.5, 241945, 251527)` $\rightarrow 1$ (rounding)



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(See log densities on the next slide)
- Probabilities extremely close to 1 may be rounded to 1
 - Ratio of girl and boy babies
 - `pbeta(0.5, 241945, 251527)` $\rightarrow 1$ (rounding)
 - `pbeta(0.5, 241945, 251527, lower.tail=FALSE)`
 $\approx 1.15 \cdot 10^{-42}$
computing the upper tail directly preserves precision



- Log densities
 - use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$

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 - Exponentiate as late as possible

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 - For $a > b$, compute
$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$

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$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$
$$\log(\exp(800) + \exp(800)) \rightarrow \text{Inf}$$



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$$\log(\exp(800) + \exp(800)) \rightarrow \text{Inf}$$
but $800 + \log(1 + \exp(800 - 800)) \approx 800.69$

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`log(exp(800) + exp(800))` $\rightarrow \text{Inf}$
but `800 + log(1 + exp(800 - 800))` ≈ 800.69
- In the Metropolis algorithm (next week), compute the log density ratio using the identity
$$\log(a/b) = \log(a) - \log(b)$$

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Pair discussion: log weights

- Suppose the log weights are $\log(w) = (-1000, -1001, -1002)$.
- Direct exponentiation may underflow. How can we obtain normalized weights without changing their relative sizes?

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After discussion:

- Subtract the largest log weight from every log weight.
- Exponentiate the shifted values and divide by their sum.
- The common shift cancels when the weights are normalized.



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Numerical calculation:

$$\begin{aligned}(-1000, -1001, -1002) - (-1000) &= (0, -1, -2), \\ \exp(0, -1, -2) &\approx (1, 0.368, 0.135), \\ \tilde{w} &\approx (0.665, 0.245, 0.090).\end{aligned}$$

The normalized weights are unchanged by the common shift.



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Section 2

Bayesian Computation



It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta) p(\theta|y) d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

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We can easily evaluate $p(y|\theta)p(\theta)$ for any θ , but the integral $\int p(y|\theta)p(\theta)d\theta$ is usually difficult.



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- Grid (equal spacing) evaluation with self-normalization

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{\sum_{s=1}^S [f(\theta^{(s)})q(\theta^{(s)}|y)]}{\sum_{s=1}^S q(\theta^{(s)}|y)}$$



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- Monte Carlo methods that can sample from $p(\theta^{(s)}|y)$ using only $q(\theta^{(s)}|y)$

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)})$$



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$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)



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 - Markov chain Monte Carlo (Ch 11-12)



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 - Numerical (deterministic) integration (grid and quadrature, Ch 3, 10)
 - Independent Monte Carlo, rejection and importance sampling (Ch 10)
 - Markov chain Monte Carlo (Ch 11-12)
 - (Distributional approximations, Ch 4, 13)



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Section 3

Monte Carlo methods



- From Lecture 3, for i.i.d. draws $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta | y)$,

$$I_h = \mathbb{E}[h(\theta) | y] = \int h(\theta) p(\theta | y) d\theta.$$

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Monte Carlo integration: recap

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$$I_h = \mathbb{E}[h(\theta) | y] = \int h(\theta)p(\theta | y) d\theta.$$

- The Monte Carlo estimator is

$$\hat{l}_S = \frac{1}{S} \sum_{s=1}^S h(\theta^{(s)}).$$



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- The Monte Carlo estimator is

$$\hat{I}_S = \frac{1}{S} \sum_{s=1}^S h(\theta^{(s)}).$$

- Unlike a grid, its evaluation points are random draws from the distribution inside the integral.



If $E[|h(\theta)| \mid y] < \infty$, then $\hat{l}_S$ is **unbiased**:

$$E[\hat{l}_S \mid y] = \frac{1}{S} \sum_{s=1}^S E[h(\theta^{(s)}) \mid y] = l_h.$$

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Unbiasedness and MCSE

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$$E[\hat{l}_S | y] = \frac{1}{S} \sum_{s=1}^S E[h(\theta^{(s)}) | y] = l_h.$$

If $\sigma_h^2 = \text{Var}[h(\theta) | y] < \infty$, independence gives

$$\text{Var}(\hat{l}_S | y) = \frac{\sigma_h^2}{S}, \quad \text{MCSE}(\hat{l}_S) = \frac{\sigma_h}{\sqrt{S}}.$$



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If $\sigma_h^2 = \text{Var}[h(\theta) | y] < \infty$, independence gives

$$\text{Var}(\hat{l}_S | y) = \frac{\sigma_h^2}{S}, \quad \text{MCSE}(\hat{l}_S) = \frac{\sigma_h}{\sqrt{S}}.$$

Estimate MCSE by replacing σ_h with the sample standard deviation of $h(\theta^{(1)}), \dots, h(\theta^{(S)})$.



- Strong law of large numbers: if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

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Monte Carlo integration: convergence

- Strong law of large numbers: if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- Mean-square convergence: if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \rightarrow 0.$$



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$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- Mean-square convergence: if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \rightarrow 0.$$

- Central limit theorem: if $0 < \sigma_h^2 < \infty$, then

$$\sqrt{S}(\hat{l}_S - l_h) \xrightarrow{d} \mathcal{N}(0, \sigma_h^2).$$



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Monte Carlo integration: convergence

- **Strong law of large numbers:** if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- **Mean-square convergence:** if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \rightarrow 0.$$

- **Central limit theorem:** if $0 < \sigma_h^2 < \infty$, then

$$\sqrt{S}(\hat{l}_S - l_h) \xrightarrow{d} \mathcal{N}(0, \sigma_h^2).$$

- Thus the typical Monte Carlo error decreases at rate $S^{-1/2}$.



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- You have $S = 1000$ independent draws.
- How many draws are needed to halve the MCSE of a posterior mean?
- Does this reduce posterior uncertainty as well?



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Pair discussion: Monte Carlo precision

- You have $S = 1000$ independent draws.
- How many draws are needed to halve the MCSE of a posterior mean?
- Does this reduce posterior uncertainty as well?

After discussion:

- MCSE decreases as $S^{-1/2}$, so four times as many draws are needed.
- We need $S = 4000$ draws.
- More draws reduce numerical error, not posterior uncertainty.



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Monte Carlo vs. deterministic integration

- In one dimension, for a sufficiently smooth function, typical error rates with N evaluations are

Method	Error rate
Midpoint or trapezoidal rule	N^{-2}
Simpson's rule	N^{-4}
Monte Carlo	$N^{-1/2}$

- Deterministic integration is therefore usually preferable for a smooth, low-dimensional problem.
Why do we still need Monte Carlo?



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- A grid with m points in each of D dimensions requires $N = m^D$ evaluations: its cost grows exponentially with D .



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- A grid with m points in each of D dimensions requires $N = m^D$ evaluations: its cost grows exponentially with D .
- The Monte Carlo rate $N^{-1/2}$ does not explicitly depend on D , which makes Monte Carlo attractive in higher dimensions.



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Pair discussion: which integration method?

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- When would you prefer deterministic quadrature?
- When would you prefer Monte Carlo integration?



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Pair discussion: which integration method?

- When would you prefer deterministic quadrature?
- When would you prefer Monte Carlo integration?

After discussion:

- Quadrature is attractive for smooth, low-dimensional integrals.
- Monte Carlo is attractive in higher dimensions and when posterior draws are already available.
- The best choice depends on both dimension and the available structure.



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Grid sampling and curse of dimensionality

- 10 parameters
- if we do not know beforehand where the posterior mass is
 - need to choose wide box for the grid
 - need to have enough grid points to get some of them in regions containing substantial posterior mass

Can we do this with a laptop (20m evals per second)?

- for example, 50 or 1000 grid points per dimension
 - $50^{10} \approx 10^{17}$ grid points
 - $1000^{10} = 10^{30}$ grid points



Grid sampling and curse of dimensionality

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- 10 parameters
- if we do not know beforehand where the posterior mass is
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 - need to have enough grid points to get some of them in regions containing substantial posterior mass
- for example, 50 or 1000 grid points per dimension
 - $50^{10} \approx 10^{17}$ grid points
 - $1000^{10} = 10^{30}$ grid points
- Even at 20 million density evaluations per second:
 - 10^{17} evaluations would take about 150 years
 - 10^{30} evaluations would take about $1.5 \cdot 10^{15}$ years



How many simulation draws are needed?

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- How many draws or how big sample size S ?
 - For independent draws
 - standard sampling formulas quantify uncertainty caused by using a finite number of draws
 - Markov chain Monte Carlo produces dependent draws (next week)
 - requires additional work to estimate the *effective sample size*



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Posterior uncertainty vs. Monte Carlo error

- **Posterior uncertainty** is the spread of $p(\theta | y)$
 - it describes what we know about θ after observing the data
 - it does not disappear when we generate more posterior draws



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- **Monte Carlo error** comes from using finitely many draws to estimate a posterior summary

$$\widehat{E(\theta | y)} = \frac{1}{S} \sum_{s=1}^S \theta^{(s)}, \quad \text{MCSE} = \frac{\sigma_{\theta}}{\sqrt{S}}.$$

- it decreases when we generate more draws



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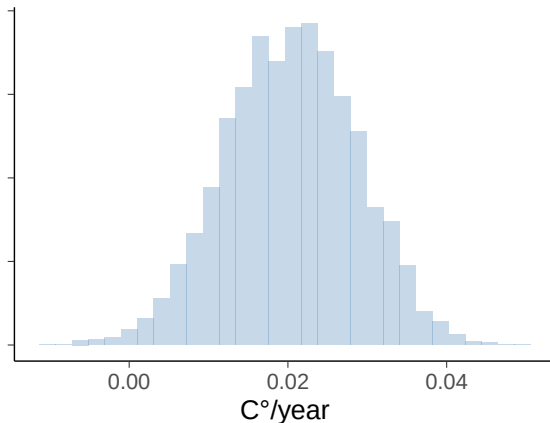
- it decreases when we generate more draws
- These are different sources of uncertainty and should be assessed separately.



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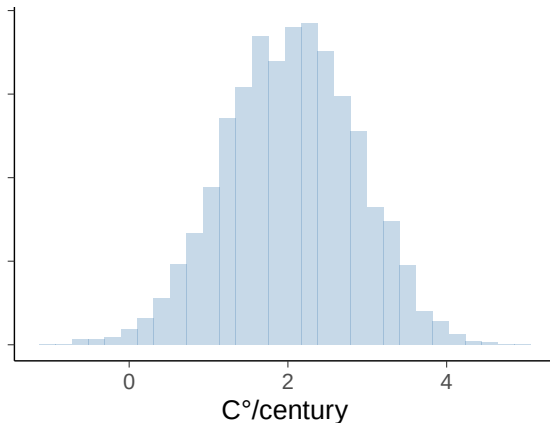
Example: Kilpisjärvi summer temperature

Posterior of temperature change





Posterior of temperature change



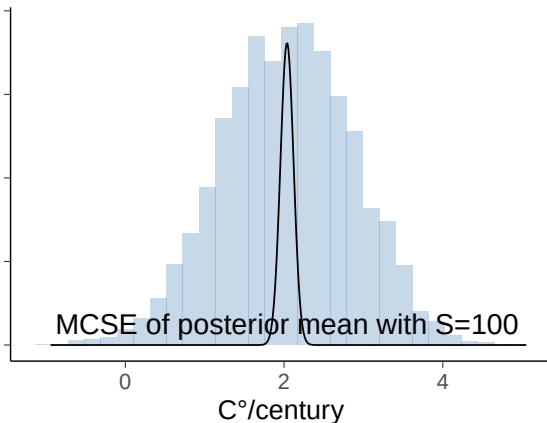
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Example: Kilpisjärvi summer temperature

Posterior of temperature change



Posterior sd ≈ 0.827 ; MCSE of the mean ≈ 0.0827

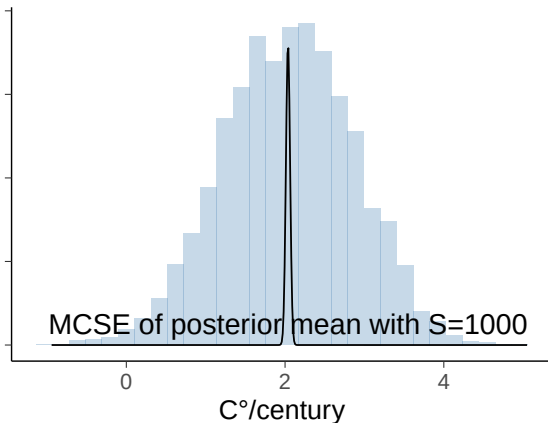
More draws reduce MCSE, but do not reduce posterior uncertainty.



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Example: Kilpisjärvi summer temperature

Posterior of temperature change



Posterior sd ≈ 0.827 ; MCSE of the mean ≈ 0.0261

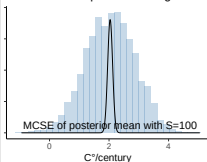
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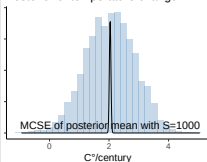
Example: Kilpisjärvi summer temperature

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Posterior of temperature change



Posterior of temperature change





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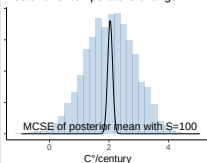
- Direct sampling

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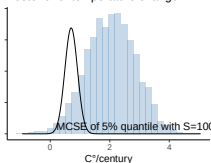
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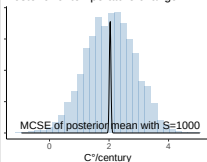
Posterior of temperature change



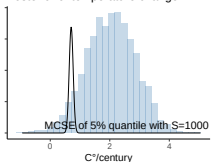
Posterior of temperature change



Posterior of temperature change



Posterior of temperature change





Example: Kilpisjärvi summer temperature

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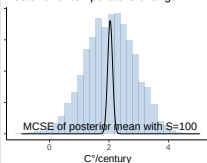
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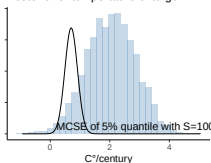
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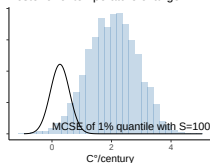
Posterior of temperature change



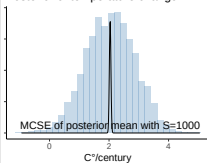
Posterior of temperature change



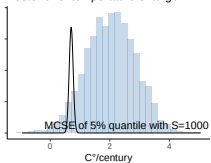
Posterior of temperature change



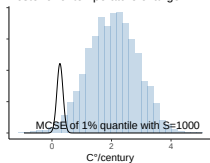
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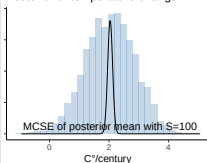
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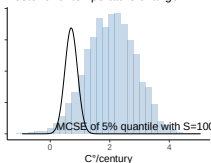
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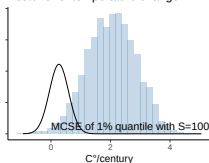
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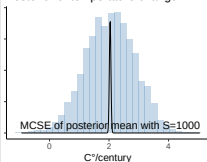
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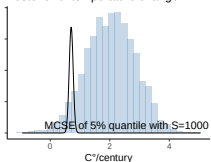
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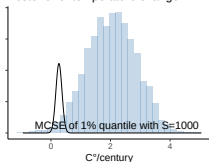
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Tail quantiles are more difficult to estimate



- Let $p_A = P(\theta \in A \mid y)$ and define $I_s = I(\theta^{(s)} \in A)$. Then

$$I_s \sim \text{Bernoulli}(p_A), \quad \hat{p}_A = \frac{1}{S} \sum_{s=1}^S I_s.$$

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Estimating a posterior probability

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- The number of draws in A is $\sum_s I_s \sim \text{Binomial}(S, p_A)$, so

$$\text{MCSE}(\hat{p}_A) = \sqrt{\frac{p_A(1-p_A)}{S}} \approx \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{S}}.$$



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- If $S = 100$ and $p_A \approx 0.5$, MCSE is 0.05, or about 5 percentage points. An MCSE of 0.01 requires about 2500 draws.

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Small probabilities require many draws

Let N_A be the number of posterior draws that fall in A . Then

$$N_A \sim \text{Binomial}(S, p_A), \quad \mathbb{E}(N_A) = Sp_A.$$

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- If $S = 1/p_A$, we expect only one draw in A . The observed count can easily be 0, 1, or 2, so the estimate is unstable.



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- If $S = 1/p_A$, we expect only one draw in A . The observed count can easily be 0, 1, or 2, so the estimate is unstable.
- For small p_A , the relative Monte Carlo error is approximately

$$\frac{\text{MCSE}(\hat{p}_A)}{p_A} \approx \frac{1}{\sqrt{Sp_A}}.$$

Thus, we need S much larger than $1/p_A$, or equivalently Sp_A much larger than 1.



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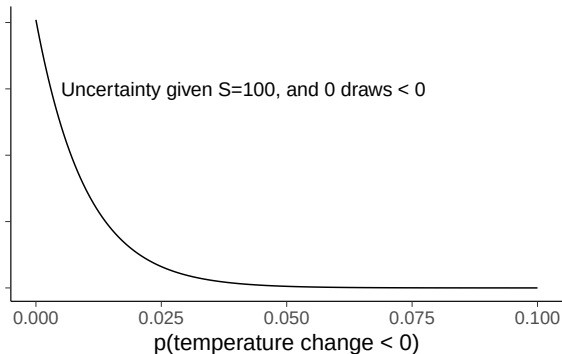
Thus, we need S much larger than $1/p_A$, or equivalently Sp_A much larger than 1.

- Example: if $p_A = 0.001$, then $S = 1000$ gives only one expected draw in A , while $S = 100\,000$ gives 100 expected draws.



Example: Kilpisjärvi summer temperature

Posterior uncertainty $p(\text{temperature change} < 0)$

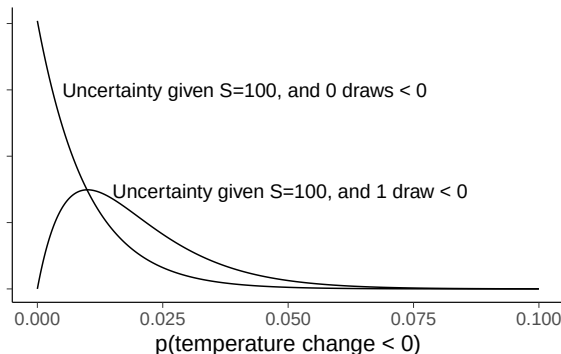


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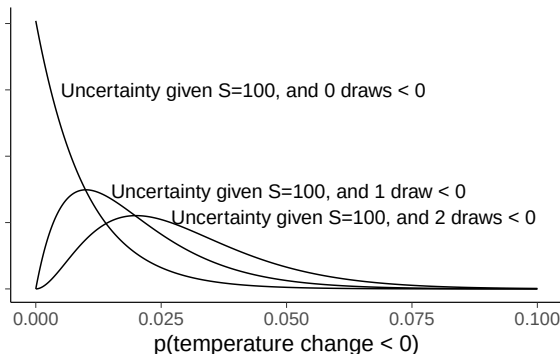


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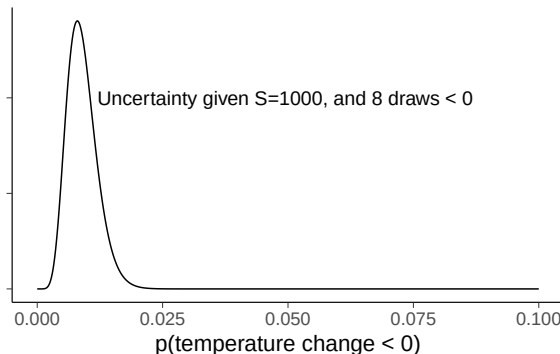


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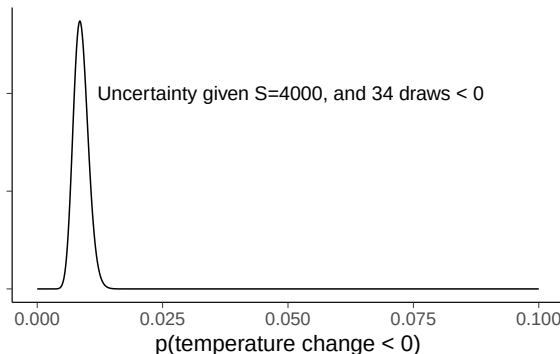


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How many digits to show in reports?

- Too many digits make reading of the results slower and give false impression of the accuracy

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- Too many digits make reading of the results slower and give false impression of the accuracy
 - Do not show digits that are just random noise
 - check what is the Monte Carlo standard error



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- Too many digits make reading of the results slower and give false impression of the accuracy
 - Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
 - Show meaningful digits given the posterior uncertainty



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- Example: The mean and 90% central posterior interval for the temperature increase in $^{\circ}\text{C}/\text{century}$, based on posterior draws



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 - 2.050774 and [0.7472868 3.3017524] (NO!)



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 - 2 and [1 3] (depends on the context)



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- Example: The probability that the temperature increase is positive



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- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)



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 - 1.00 (depends on the context)



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- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)
 - 1.00 (depends on the context)
 - With 4000 draws $MCSE \approx 0.002$. We could report that the probability is **very likely larger than 0.99**, or sample more to justify reporting three digits



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 - For probabilities close to 0 or 1, also consider whether the model assumptions justify such precision



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 - 0.9960000 (NO!)
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 - With 4000 draws $MCSE \approx 0.002$. We could report that the probability is **very likely larger than 0.99**, or sample more to justify reporting three digits
 - For probabilities close to 0 or 1, also consider whether the model assumptions justify such precision
- **For your project:** For each reported value, choose a sensible number of digits.



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 - **Direct sampling**
 - Indirect sampling
 - Rejection sampling
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- Direct simulation from a known density or mass function, such as $p(\theta | y)$ in a conjugate model
 - Produces independent draws
 - analytic transformations of uniform random numbers (e.g. Appendix A)
 - factorization
 - numerical inversion of the CDF
 - **Limitation**: available only for a limited set of models



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Section 5

Indirect sampling



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- Rejection sampling
- Importance sampling
- Markov chain Monte Carlo (next week)



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- The effective sample size S_{eff} compares Monte Carlo precision with that of independent, equally weighted draws
 - For S independent draws, $S_{\text{eff}} = S$



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- The effective sample size S_{eff} compares Monte Carlo precision with that of independent, equally weighted draws
- For S independent draws, $S_{\text{eff}} = S$
- Weighted or correlated draws usually have $S_{\text{eff}} < S$
- A larger S_{eff} means more precise estimates for the same number of stored draws



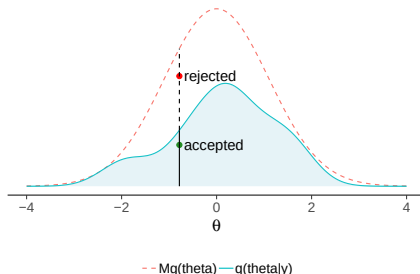
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Rejection sampling

- Let $p(\theta) = q(\theta)/Z$ be the target density
- Choose a proposal g and a finite M such that

$$q(\theta) \leq Mg(\theta) \quad \text{for all } \theta$$

- Intuitively, scale g vertically by M until Mg covers q everywhere. Then $q(\theta)/(Mg(\theta))$ is always between 0 and 1.





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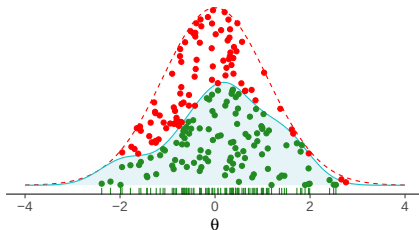
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- Draw $\theta^* \sim g$ and $u \sim \mathcal{U}(0, 1)$; accept if

$$u \leq \frac{q(\theta^*)}{Mg(\theta^*)}$$



• Accepted • Rejected - - Mg(theta) — q(theta)



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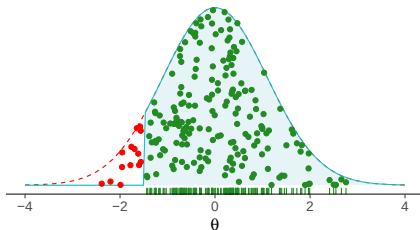
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Why rejection sampling works

- The density of an accepted proposal is proportional to

$$\underbrace{g(\theta)}_{\text{proposal density}} \cdot \underbrace{\frac{q(\theta)}{Mg(\theta)}}_{\text{acceptance probability}} = \frac{q(\theta)}{M} \propto p(\theta).$$



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- More formally, for any region B ,

$$P(\theta \in B \mid \text{accepted}) = \frac{\int_B q(\theta)/M \, d\theta}{\int q(\theta)/M \, d\theta} = \int_B p(\theta) \, d\theta.$$

Thus, accepted values are independent draws from the target.



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- If A of T proposals are accepted, their effective sample size is A , while A/T estimates the acceptance rate Z/M
When will the acceptance rate be high or low?



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- If A of T proposals are accepted, their effective sample size is A , while A/T estimates the acceptance rate Z/M
 - A tight envelope gives a high acceptance rate
 - Finding a good envelope becomes difficult as dimension increases



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Importance sampling

If we can evaluate the normalized target p and have a proposal g , then we can use **importance sampling**

$$w(\theta) = \frac{p(\theta)}{g(\theta)},$$

$$\mathbb{E}_p[f(\theta)] = \int f(\theta)w(\theta)g(\theta) d\theta = \mathbb{E}_g[w(\theta)f(\theta)].$$



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$$\mathbb{E}_p[f(\theta)] = \int f(\theta) w(\theta) g(\theta) d\theta = \mathbb{E}_g[w(\theta) f(\theta)].$$

Thus, for $\theta^{(s)} \sim g$,

$$\hat{l}_S = \frac{1}{S} \sum_{s=1}^S w(\theta^{(s)}) f(\theta^{(s)})$$

is an unbiased estimator of $\mathbb{E}_p[f(\theta)]$.



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Importance sampling

If we can evaluate the normalized target p and have a proposal g , then we can use **importance sampling**

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is an unbiased estimator of $\mathbb{E}_p[f(\theta)]$. If only $q(\theta) = Zp(\theta)$ is known, the unknown Z cancels:

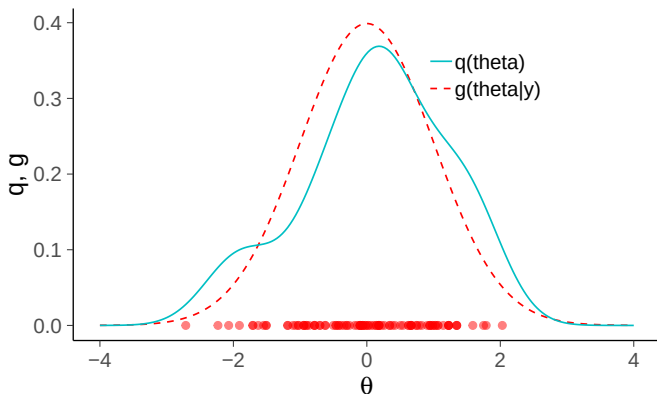
$$\mathbb{E}_p[f(\theta)] = \frac{\mathbb{E}_g[(q/g)f]}{\mathbb{E}_g[q/g]}.$$



Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Target, proposal, and draws



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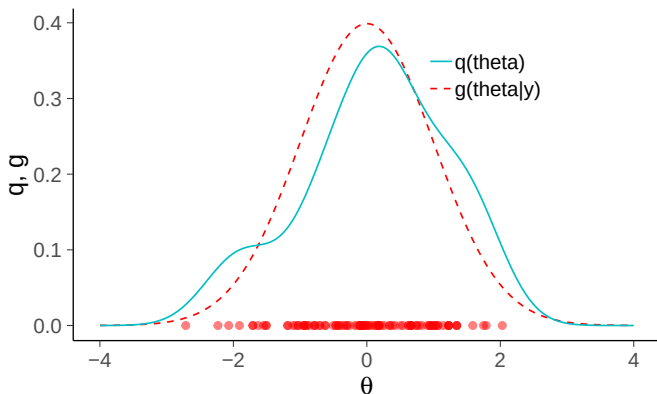


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Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Target, proposal, and draws



$$\widehat{E_p[f(\theta)]} = \sum_{s=1}^S \tilde{w}_s f(\theta^{(s)}), \quad \tilde{w}_s = \frac{w_s}{\sum_{r=1}^S w_r}$$

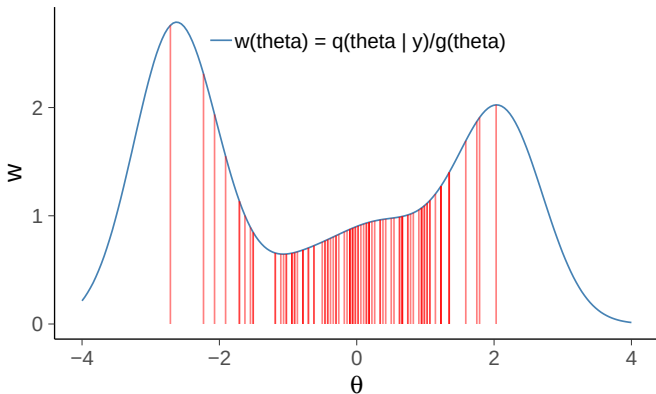


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Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Draws and importance weights



$$\widehat{E_p[f(\theta)]} = \sum_{s=1}^S \tilde{w}_s f(\theta^{(s)}), \quad \tilde{w}_s = \frac{w_s}{\sum_{r=1}^S w_r}$$

The estimator is generally biased for finite S , but consistent as $S \rightarrow \infty$.



Pair discussion: proposal tails

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- The target has heavier tails than the proposal.
- What happens to the importance weights in the tails?
- Would a slightly wider proposal be safer or less safe?



Pair discussion: proposal tails

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- The target has heavier tails than the proposal.
- What happens to the importance weights in the tails?
- Would a slightly wider proposal be safer or less safe?

After discussion:

- Rare tail draws receive extremely large weights.
- A few draws may dominate, producing low ESS and unstable estimates.
- A proposal with sufficiently heavy tails is usually safer.



Choosing a proposal

- The proposal must satisfy

$$g(\theta) > 0 \quad \text{wherever } p(\theta) > 0$$

Otherwise, important parts of the target can never be sampled.

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Choosing a proposal

- The proposal must satisfy

$$g(\theta) > 0 \quad \text{wherever } p(\theta) > 0$$

Otherwise, important parts of the target can never be sampled.

- A good proposal covers the target and gives similar weights
 - tails that are too light can produce rare, extremely large weights
 - large weight variation gives unstable estimates and low effective sample size

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Weight variability

- Variation in the weights affects the **effective sample size**
 - if one weight dominates, we have effectively one draw
 - if all weights are equal, we have effectively S draws

Which weight pattern would you trust more?



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- Variation in the weights affects the **effective sample size**
 - if one weight dominates, we have effectively one draw
 - if all weights are equal, we have effectively S draws
- For the usual central limit theorem, the weighted quantity must have finite variance; for self-normalized importance sampling this requires

$$\mathbb{E}_g \left[w(\theta)^2 (f(\theta) - \mathbb{E}_p[f(\theta)])^2 \right] < \infty.$$



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$$E_g \left[w(\theta)^2 (f(\theta) - E_p[f(\theta)])^2 \right] < \infty.$$

- Heavy-tailed weights can violate this condition even when $f(\theta)$ itself has finite variance under the target.



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1. Draw $(\alpha^{(s)}, \beta^{(s)})$ from the bivariate normal proposal g
2. Compute log weights
 $\log w_s = \log q(\alpha^{(s)}, \beta^{(s)}) - \log g(\alpha^{(s)}, \beta^{(s)})$
3. Stabilize and normalize the weights so that $\sum_s \tilde{w}_s = 1$
4. Compute weighted posterior summaries and inspect the weights and S_{eff}
5. Optionally resample draws with probabilities $\tilde{w}_s$



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Bioassay importance-sampling workflow

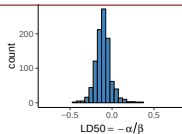
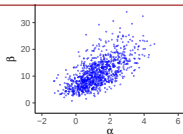
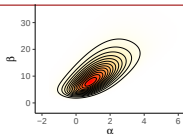
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IR in the following figures means **importance resampling**:
resampling the proposal draws using the normalized weights.



Bioassay: proposal and resampled draws

Grid



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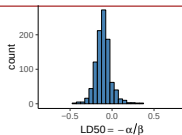
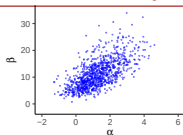
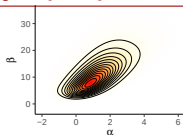


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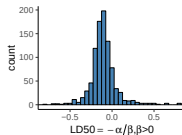
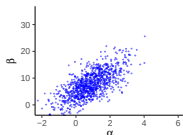
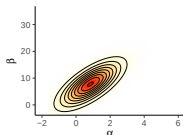
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Bioassay: proposal and resampled draws

Grid



Normal

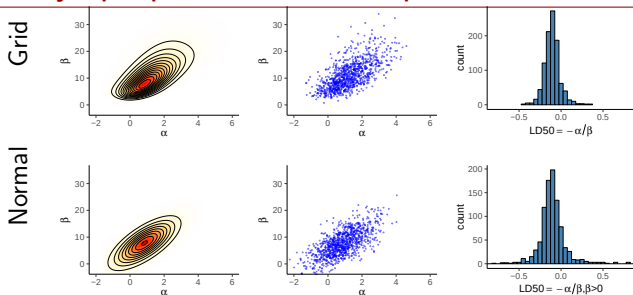


Normal approximation is discussed more in BDA3 Ch 4



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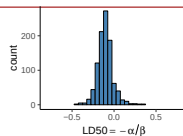
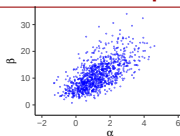
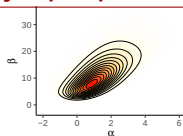


Normal approximation is discussed more in BDA3 Ch 4
But the normal approximation is not accurate here:
 $\text{Grid SD}(\text{LD50}) \approx 0.1$, $\text{normal SD}(\text{LD50}) \approx 0.75$

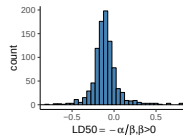
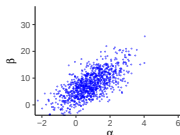
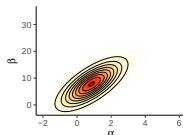


Bioassay: proposal and resampled draws

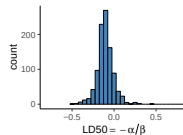
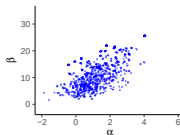
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Normal



IR



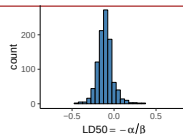
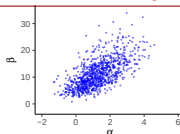
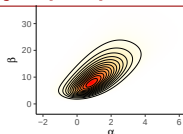
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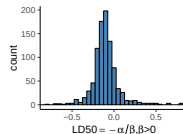
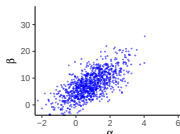
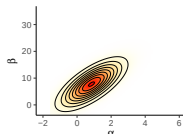
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Bioassay: proposal and resampled draws

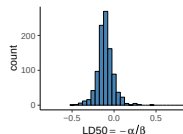
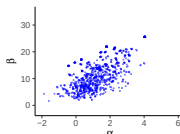
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Normal



IR



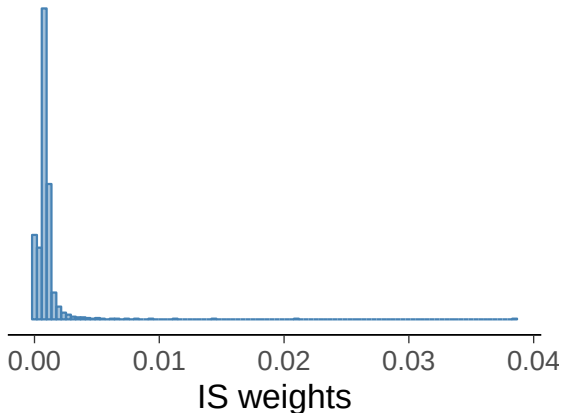
Grid $SD(LD50) \approx 0.1$, IR $SD(LD50) \approx 0.1$



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Bioassay: importance weights

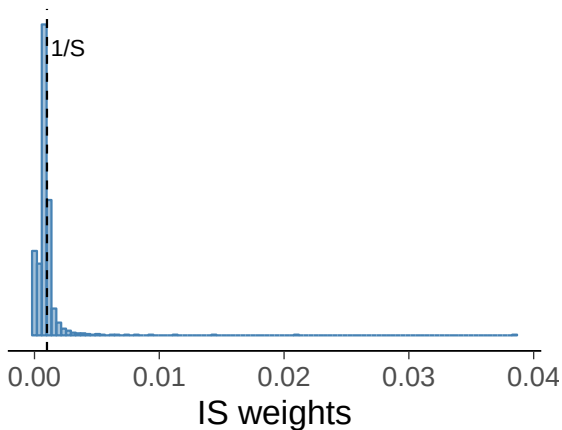




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Bioassay: importance weights





Pair discussion: can we trust the weights?

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- Most normalized weights are near zero, while a few are much larger.
 - What does this imply about the estimate?



Pair discussion: can we trust the weights?

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- What does this imply about the estimate?

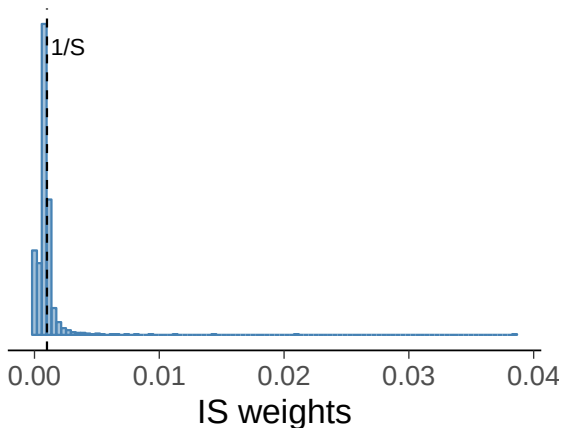
After discussion:

- Only a small number of draws contribute substantial information.
- The estimate may be sensitive to a few draws.



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Bioassay: effective sample size

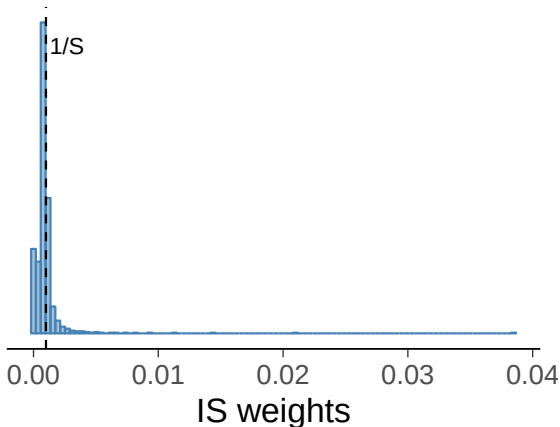


$$S_{\text{eff}} = \frac{1}{\sum_{s=1}^S (\tilde{w}(\theta^s))^2}, \quad \text{where } \tilde{w}(\theta^s) = w(\theta^s) / \sum_{s'=1}^S w(\theta^{s'})$$



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Bioassay: effective sample size



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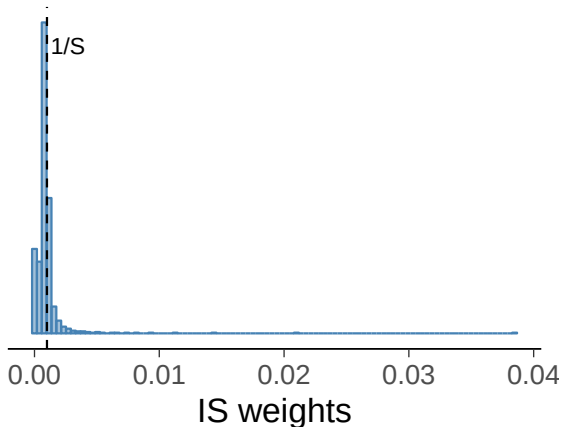
BDA3 1st (2013) and 2nd (2014) printing have an error for $\tilde{w}(\theta^s)$. The normalized weights equation should not have the multiplier S (the normalized weights should sum to one). Errata for the book

http://www.stat.columbia.edu/~gelman/book/errata_bda3.txt



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Bioassay: effective sample size



$$S_{\text{eff}} = \frac{1}{\sum_{s=1}^S (\tilde{w}(\theta^s))^2}, \quad \text{where } \tilde{w}(\theta^s) = w(\theta^s) / \sum_{s'=1}^S w(\theta^{s'})$$
$$S_{\text{eff}} \approx 270$$



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Pareto-smoothed importance sampling

- Raw importance sampling can be dominated by a few very large weights

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- Raw importance sampling can be dominated by a few very large weights
- PSIS fits a generalized Pareto distribution to the upper tail of the importance ratios
- It replaces the largest ratios with smoothed values from the fitted tail



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Pareto-smoothed importance sampling

- Raw importance sampling can be dominated by a few very large weights
- PSIS fits a generalized Pareto distribution to the upper tail of the importance ratios
- It replaces the largest ratios with smoothed values from the fitted tail
- This usually reduces variability, at the cost of a small amount of bias
- PSIS also provides improved estimates of effective sample size and MCSE

Vehtari, Simpson, Gelman, Yao, and Gabry (2024)



The Pareto $\hat{k}$ diagnostic

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- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
 - smaller $\hat{k}$ means more stable importance sampling
 - larger $\hat{k}$ means that a few ratios dominate



The Pareto $\hat{k}$ diagnostic

- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
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- Current guidance treats estimates as reliable when

$$\hat{k} < 0.7$$

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- Above this threshold, the estimate may be unreliable; investigate the proposal or use a more robust computation

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The Pareto $\hat{k}$ diagnostic

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- Above this threshold, the estimate may be unreliable; investigate the proposal or use a more robust computation
- For intuition: a generalized Pareto tail has finite variance when $k < 1/2$; when $k \geq 1$, even its mean does not exist



Importance-sampling leave-one-out cross-validation

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- Later in the course, we use $p(\theta | y)$ as a proposal distribution for $p(\theta | y_{-i})$
 - this allows fast approximation of leave-one-out cross-validation

$$p(y_i | y_{-i}) = \int p(y_i | \theta) p(\theta | y_{-i}) d\theta$$



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Recap: simulation methods

Method	Draw from	Output	Main limitation
Direct	Target p	Independent, equal-weight draws	Available for few models
Rejection	Proposal g	Accepted independent target draws	Acceptance can be very low
Importance	Proposal g	Weighted proposal draws	A few weights may dominate
MCMC	Transition rule	Dependent target draws	Requires convergence diagnostics



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Importance	Proposal g	Weighted proposal draws	A few weights may dominate
MCMC	Transition rule	Dependent target draws	Requires convergence diagnostics

- There is no universally best method: efficiency depends on the target, dimension, proposal, and quantity being estimated.
- Always assess Monte Carlo accuracy with appropriate diagnostics.



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Section 6

Additional material



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Riemann recap

Partition $[a, b]$ as

$$a = x_0 < x_1 < \cdots < x_S = b, \quad \Delta x_s = x_s - x_{s-1},$$

and choose one point $x_s^* \in [x_{s-1}, x_s]$ in each subinterval. The corresponding **Riemann sum** is

$$R_S = \sum_{s=1}^S f(x_s^*) \Delta x_s.$$



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$$R_S = \sum_{s=1}^S f(x_s^*) \Delta x_s.$$

The **Riemann integral** is the limit

$$\int_a^b f(x) dx = \lim_{\max_s \Delta x_s \rightarrow 0} \sum_{s=1}^S f(x_s^*) \Delta x_s.$$



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Numerical integration

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- Use a finite Riemann sum to approximate the integral:



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- Use a finite Riemann sum to approximate the integral:

$$\int_a^b f(x) dx \approx \sum_{s=1}^S f(x_s^*) \Delta x_s,$$

- For S equal-width subintervals, $\Delta x = (b - a)/S$.
- A common choice is the **midpoint**:

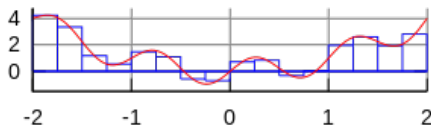
$$x_s^* = \frac{x_{s-1} + x_s}{2} = a + \left(s - \frac{1}{2}\right) \Delta x, \quad s = 1, \dots, S.$$



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- Midpoint rule

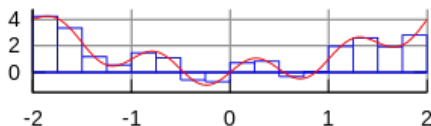




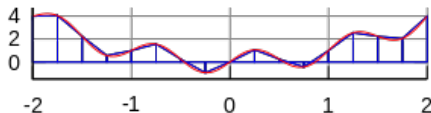
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Numerical integration

- Midpoint rule



- Another rule with smaller error: trapezoidal rule

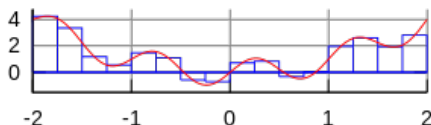




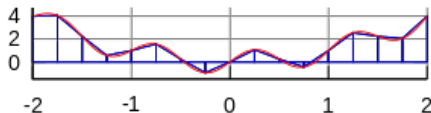
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- Another rule with smaller error: trapezoidal rule



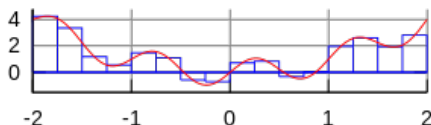
- Higher-order approaches also exist, such as Simpson's rule



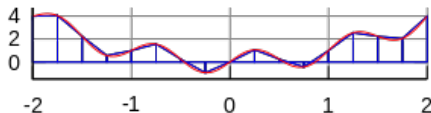
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Numerical integration

- Midpoint rule



- Another rule with smaller error: trapezoidal rule



- Higher-order approaches also exist, such as Simpson's rule
- But there is a curse of dimensionality: S points per dimension require S^D grid points in D dimensions



- Used already before computers
 - Buffon (18th century; needles)
 - De Forest, Darwin, Galton (19th century)
 - Pearson (19th century; roulette)
 - Gosset (Student, 1908; hat)

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Monte Carlo: history

- Used already before computers
 - Buffon (18th century; needles)
 - De Forest, Darwin, Galton (19th century)
 - Pearson (19th century; roulette)
 - Gosset (Student, 1908; hat)
- The term “Monte Carlo method” emerged from work by Metropolis, von Neumann, and Ulam in the late 1940s
 - they worked together on the Manhattan Project
 - Metropolis and Ulam, “The Monte Carlo Method”, 1949



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- The term “Monte Carlo method” emerged from work by Metropolis, von Neumann, and Ulam in the late 1940s
 - they worked together on the Manhattan Project
 - Metropolis and Ulam, “The Monte Carlo Method”, 1949
- Monte Carlo methods became widely practical for Bayesian inference as computing became cheaper in the 1990s
 - BUGS project started 1989 (last OpenBUGS release 2014)
 - Gelfand & Smith, 1990
 - Stan initial release 2012



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- Simulate draws from the target distribution $p(\theta|y)$
 - these draws can be analyzed like a random sample
 - a collection of draws has sample size S



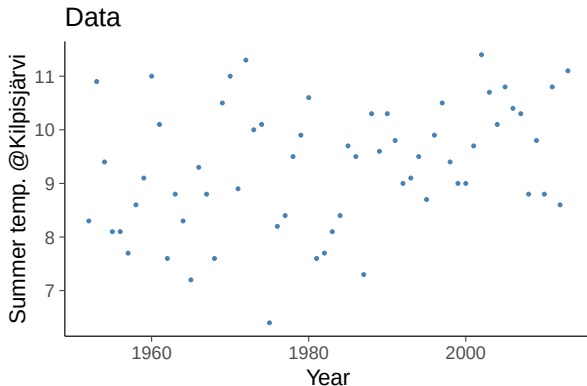
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- Simulate draws from the target distribution $p(\theta|y)$
 - these draws can be analyzed like a random sample
 - a collection of draws has sample size S
 - Use these draws, for example,
 - to compute means, standard deviations, and quantiles
 - to draw histograms
 - to marginalize
 - to estimate probabilities and derived quantities



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Example: Kilpisjärvi summer temperature

Average temperature in June, July, and August at Kilpisjärvi, Finland

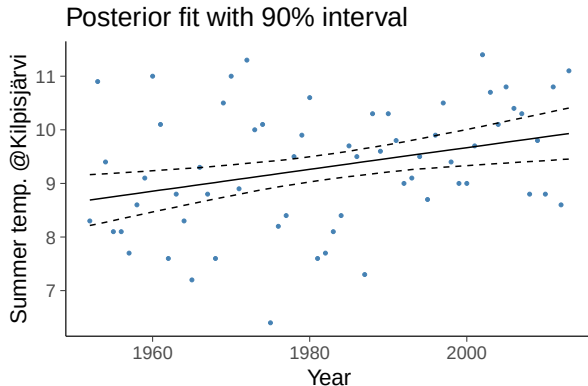




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Example: Kilpisjärvi summer temperature

Average temperature in June, July, and August at Kilpisjärvi, Finland





How many simulation draws are needed?

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- Fewer draws may be needed with
 - deterministic methods
 - Rao-Blackwellization (marginalisation)
 - variance-reduction methods, such as control variates



How many simulation draws are needed?

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- The $S^{-1/2}$ Monte Carlo rate does not explicitly depend on dimension



How many simulation draws are needed?

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- The $S^{-1/2}$ Monte Carlo rate does not explicitly depend on dimension
 - However, obtaining useful independent draws can be difficult in high-dimensional problems



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Random number generators

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- How do we sample from a probability distribution?



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- How do we sample from a probability distribution?
 - Good **pseudorandom** number generators are sufficient for Bayesian inference
 - a pseudorandom generator uses a deterministic algorithm to produce a sequence that behaves like a random sequence
 - modern statistical software uses high-quality generators



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- Box–Muller method:

If U_1 and U_2 are independent draws from $\mathcal{U}(0, 1)$, and

$$X_1 = \sqrt{-2 \log(U_1)} \cos(2\pi U_2)$$

$$X_2 = \sqrt{-2 \log(U_1)} \sin(2\pi U_2)$$

then X_1 and X_2 are independent draws from the distribution $\mathcal{N}(0, 1)$



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$$X_2 = \sqrt{-2 \log(U_1)} \sin(2\pi U_2)$$

then X_1 and X_2 are independent draws from the distribution $\mathcal{N}(0, 1)$

- it is not the fastest method because it requires trigonometric functions
- many alternative methods exist for the normal distribution
- R uses numerical inversion by default



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Pair discussion: rejection proposal

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- What happens if Mg fails to cover the target everywhere?
- What happens if the envelope is valid but much wider than the target?



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Pair discussion: rejection proposal

- What happens if Mg fails to cover the target everywhere?
- What happens if the envelope is valid but much wider than the target?

After discussion:

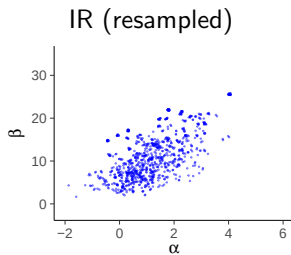
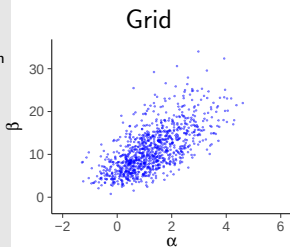
- Without full coverage, the algorithm is invalid and misses part of the target.
- An overly wide envelope is valid but gives a low acceptance rate.
- A good envelope covers the target as tightly as possible.



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Bioassay: grid and resampled draws

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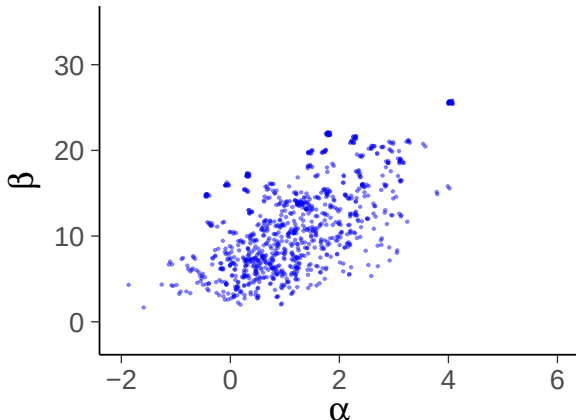


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Bioassay: importance-resampled draws

IR (resampled)





Next week: Markov chain Monte Carlo (MCMC)

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 - Additional material
- Pros
 - Markov chain goes where most of the posterior mass is
 - Certain MCMC methods scale well to high dimensions
 - Cons
 - Draws are dependent (affects how many draws are needed)
 - Convergence in practical time is not guaranteed



Next week: Markov chain Monte Carlo (MCMC)

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 - Markov chain goes where most of the posterior mass is
 - Certain MCMC methods scale well to high dimensions
- Cons
 - Draws are dependent (affects how many draws are needed)
 - Convergence in practical time is not guaranteed
- MCMC methods in this course
 - Gibbs sampling: “iterative conditional sampling”
 - Metropolis: “random walk in joint distribution”
 - Dynamic Hamiltonian Monte Carlo: “state-of-the-art” used in Stan